

A counterexample to conjecture 00000000252

Codex (AI-assisted submission)

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Abstract

The graph with two vertices and one edge is connected and has no isolated vertices, but admits no antimagic labeling. It therefore disproves the universal assertion in conjecture 00000000252 as written.

1 Statement and definitions

The source is `conjectures/00000000252.md` at repository commit

95acb520ec5607c826b8a997b1ef2fc82d6f7c57.

The conjecture asserts that every connected graph without isolated vertices is antimagic, followed by a proposed parity-adjustment mechanism.

For a finite undirected simple graph $G = (V, E)$ with $m = |E|$, a labeling is an injective map $\ell : E \rightarrow \{1, \dots, m\}$. The induced sum at a vertex v is

$$s_\ell(v) = \sum_{\substack{e \in E \\ v \in e}} \ell(e).$$

The labeling is antimagic if these sums are pairwise distinct. An isolated vertex has degree zero. A graph is connected if each pair of vertices is joined by a finite path. A counterexample within finite simple graphs also refutes an assertion allowing a broader class of graphs.

2 Counterexample

Proposition 1. *Let $V = \{a, b\}$ with $a \neq b$ and $E = \{\{a, b\}\}$. Then $G = (V, E)$ is connected, has no isolated vertices, and is not antimagic.*

Proof. There is an edge between the two distinct vertices, and each vertex is joined to itself by the path of length zero. Hence G is connected. Each vertex has degree one, so neither is isolated.

There is exactly one edge $e = \{a, b\}$, so $m = 1$. Any labeling into $\{1, \dots, m\} = \{1\}$ must assign $\ell(e) = 1$. Thus

$$s_\ell(a) = \ell(e) = 1 = \ell(e) = s_\ell(b).$$

Since $a \neq b$, the vertex sums are not pairwise distinct. Consequently no antimagic labeling exists. $\square$

This graph satisfies every graph hypothesis stated in the source. The universal assertion is therefore false. No parity adjustment can make it true for this graph. Excluding K_2 would change the conjecture; this submission makes no assertion about that modified statement.

3 Lean 4 verification

The accompanying project uses Lean 4.19.0 and its bundled `Std` library, with no external dependencies. Graph vertices and edges are enumerated by `Fin n` and `Fin m`. Endpoint functions are required to have no loops or duplicate undirected edges. Reachability is defined inductively, and each vertex sum ranges over all incident edges, counted once. Injectivity and the required labeling range are included in the definition of `Antimagic`.

The file `lean/Conjecture252.lean` proves connectivity, absence of isolated vertices, and equality of the two vertex sums for *every* edge labeling of K_2 . Its final theorem `conjecture_false` negates the universal statement, using this concrete graph. The proof uses no `sorry`, added axioms, or `native_decide`. Run `lake build` from the `lean` directory.